

Inertial Measurement Unit

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What is an IMU?

An Inertial Measurement Unit (IMU) is a self-contained electronic device that measures a body's linear acceleration, angular velocity and—when a magnetometer is included—the surrounding magnetic field. The smallest “6-axis” IMU contains a 3-axis accelerometer and a 3-axis gyroscope; a “9-axis” unit adds a 3-axis magnetometer. By integrating these data an IMU allows a CPU to compute 3-D orientation, velocity changes and, for short intervals, relative position, all without external infrastructure such as GPS or optical references.

Working Principle

Angular velocity is sensed by gyroscopes (Coriolis or Sagnac effect). Specific force (acceleration minus gravity) is sensed by accelerometers (mass-spring or MEMS capacitive principle). Raw readings are time-integrated to obtain delta-angle and delta-velocity increments. A micro-processor fuses the increments, compensates for temperature drift, bias and scale-factor errors, and outputs orientation increments or full attitude/heading solution at rates up to 1 kHz.

Core Components

- Accelerometers – MEMS, quartz, piezo-electric or optical; measure specific force along X-Y-Z.
- Gyroscopes – MEMS, Fibre-Optic (FOG) or Ring-Laser (RLG); measure angular rate about X-Y-Z.
- Magnetometer (optional) – Hall or flux-gate; delivers Earth-field vector for absolute heading.
- Temperature sensor – permits thermal compensation.
- MCU / DSP – runs calibration, alignment and fusion algorithms.

Types of IMUs

Consumer MEMS – Silicon MEMS, 10–100 °/h bias, phones & game controllers.

Industrial MEMS – Quartz MEMS, 1–10 °/h bias, AGVs & drones.

Tactical FOG – Fibre-optic, 0.1–1 °/h bias, UAVs & missiles.

Navigation RLG – Ring-laser, 0.001–0.01 °/h bias, airliners & spacecraft.

Importance of IMUs

- GPS-denied navigation: tunnels, indoor, underwater, planetary surface.
- High-rate feedback: 100–1000 Hz updates for vehicle stability, camera gimbals, VR tracking.
- Dead-reckoning: bridges GPS outages in rovers, cars, aircraft.
- Small & power-efficient: enables swarm robotics, wearables, IoT.

Quaternions & Euler Angles

Euler Angles – roll (ϕ), pitch (θ), yaw (ψ): intuitive but suffer gimbal lock when pitch = $\pm 90^\circ$.

Quaternions – $q = [q_0, q_1, q_2, q_3]$, a 4-D unit vector; singularity-free, preferred for integrating gyro data and for 3-D graphics/robotics. Conversion routines shuttle between DCM $\leftrightarrow$ quaternion $\leftrightarrow$ Euler depending on human readability or algorithmic needs.

Sensor Fusion

Combines redundant, complementary sensors to outperform any single sensor. Algorithms: Extended Kalman Filter (EKF), Unscented KF, Particle filters, complementary filter.

Applications: attitude-heading reference systems (AHRS), visual-inertial SLAM, rover tilt safety & wheel-slip detection, smartphone image stabilisation.

Challenges Faced by IMUs on Rover

Temperature swings on Mars (-80°C night, $+20^\circ\text{C}$ day) $\rightarrow$ time-varying bias; needs in-situ thermal calibration.

Vibration & shock from wheel-terrain interaction $\rightarrow$ accelerometer noise, gyro g-sensitivity; requires isolation foam & vibration-resistant FOG/RLG or high-rate MEMS with anti-alias filtering.

Dust accumulation on IMU housing alters thermal paths; needs sealed packaging.

Drift accumulation – no GPS aiding for long traverses; relies on periodic “stop & zero-update” or wheel-encoder / visual alignment to bound position error.

Power & mass budget – rover electronics constrained; favours low-SWaP MEMS but must still meet $\leq 0.1^\circ$ attitude knowledge for mast pointing & high-gain antenna steering.